Tech Dessert #8: Fixing Interference and Signal Problems

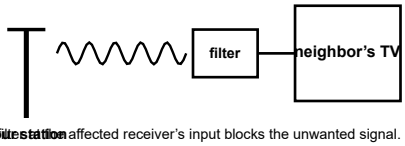
Companion to Bite #8 -- what causes interference, and the calm, correct steps to track it down and fix it.

Group T7B · 11 pool questions · Self-study handout · Practice exams at eham.net/exams

This Dessert is the companion to Bite 8 The Bite covered what is inside a radio; this sheet covers what to do when a signal goes wrong – yours interfering with someone, or something interfering with you. Read the table, study the diagram, then drill the questions.

Tracking Down Interference

Situation	What to do / what to know	Q
Your FM signal is "over-deviating"	Talk **farther from the microphone**.	T7B01
A broadcast radio picks up your ham signal	The broadcast receiver **cannot reject strong signals** from outside its band.	T7B02
What causes radio-frequency interference	Fundamental overload, harmonics, and spurious emissions – **all of these**.	T7B03
Low RF power output from a solid-state radio	**High SWR.**	T7B04
Reducing interference to a neighbor's radio	Put a **filter at the antenna input of the affected receiver**.	T7B05
A neighbor says you interfere with their TV	First make sure your station **works properly** and does not interfere with your own TV on that channel.	T7B06
Interference to your 2 m radio from a commercial FM station	Install a **band-reject filter**.	T7B07
Something in a neighbor's home interferes with you	Work with the neighbor, inform them, and keep your own station to good practice – **all of these**.	T7B08
First step for cable-TV interference from your signal	Make sure all the TV's **coax connectors are installed properly**.	T7B09
Your audio through an FM repeater is distorted	Off frequency, talking too loud or too close, or a bad location – **any of these**.	T7B10
Eliminating distorted voice transmissions	Add a clip-on **ferrite choke** to the microphone cable.	T7B11



affected receiver's input blocks the unwanted signal.

To stop your signal from interfering with a neighbor's device, the filter goes at the input of the affected receiver – not on your transmitter.

Remember: interference is usually fixed at the **receiver that has the problem**, not the transmitter – a broadcast radio picking up a ham signal simply cannot reject strong nearby signals, so a **filter at that receiver** is the cure. If a neighbor complains, stay calm: first prove your **own station is clean**, then help them. And a **ferrite choke** on the microphone cable cures distorted audio caused by RF feedback.

Dessert 8 Question Bank (T7B)

T7B01. What can you do if you are told your FM handheld or mobile transceiver is over-deviating?
A. Talk louder into the microphone
B. Let the transceiver cool off
C. Change to a higher power level
D. Talk farther away from the microphone

T7B02. What would cause a broadcast AM or FM radio to receive an amateur radio transmission unintentionally?
A. The receiver is unable to reject strong signals outside the AM or FM band
B. The microphone gain of the transmitter is turned up too high
C. The audio amplifier of the transmitter is overloaded
D. The deviation of an FM transmitter is set too low

T7B03. Which of the following can cause radio frequency interference?
A. Fundamental overload
B. Harmonics
C. Spurious emissions
D. All these choices are correct

T7B04. Which of the following might be the cause of low RF power output from a solid-state transceiver?
A. Poor amplifier noise figure
B. Poor amplifier linearity
C. Low SWR
D. High SWR

T7B05. Which of the following might reduce interference by an amateur station to a non-amateur over-the-air radio receiver?
A. Block the amateur signal with a filter at the antenna input of the affected receiver
B. Block the interfering signal with a filter on the amateur transmitter
C. Switch the transmitter from FM to SSB
D. Switch the transmitter to a narrow-band mode

T7B06. Which of the following actions should you take if a neighbor tells you that your station's transmissions are interfering with their radio or TV reception?
A. Make sure that your station is functioning properly and that it does not cause interference to your own radio or television when it is tuned to the same channel
B. Immediately turn off your transmitter and contact the nearest FCC office for assistance
C. Install a harmonic doubler on the output of your transmitter and tune it until the interference is eliminated
D. All these choices are correct

T7B07. Which of the following can reduce interference to a 2-meter band transceiver from a nearby commercial FM station?
A. Installing an RF preamplifier
B. Using double-shielded coaxial cable
C. Installing bypass capacitors on the microphone cable
D. Installing a band-reject filter

T7B08. What should you do if something in a neighbor's home is causing harmful interference to your amateur station?
A. Work with your neighbor to identify the offending device
B. Politely inform your neighbor that FCC rules prohibit the use of devices that cause interference
C. Make sure your station meets the standards of good amateur practice
D. All these choices are correct

T7B09. What should be the first step to resolve non-fiber optic cable TV interference caused by your amateur radio transmission?
A. Add a low-pass filter to the TV antenna input
B. Add a high-pass filter to the TV antenna input
C. Add a preamplifier to the TV antenna input
D. Be sure all TV feed line coaxial connectors are installed properly

T7B10. What might be a problem if you receive a report that your audio signal through an FM repeater is distorted or unintelligible?
A. Your transmitter is slightly off frequency
B. You are speaking too loudly or too close to the microphone
C. You are in a bad location
D. All these choices are correct

T7B11. Which of the following can eliminate distorted voice transmissions?
A. Adding extra feedline to the antenna to lower SWR
B. Turning the radio on and off to reset the computer-controlled circuitry
C. Adding a clip-on ferrite "choke" to the microphone cable to prevent the transmitted signal from feeding back into the transmitter
D. Turning the squelch control fully clockwise to prevent the transmitted signal from triggering the squelch circuit

Answer Key

T7B01: D T7B02: A T7B03: D T7B04: D

T7B05: A T7B06: A T7B07: D T7B08: D

T7B09: D T7B10: D T7B11: C